STT 843: Multivariate Analysis

8. Factor Analysis and Inference for Structured Covariance Matrices (Chapter 9.1-9.3)

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Outline

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Factor Analysis and Inference for Structured Covariance Matrices

- The essential purpose of factor analysis is to describe, if possible, the covariance relationships among many variables in terms of a few underlying, but unobservable, random quantities called factors
- Factor analysis can be considered an extension of *principal component analysis*. Both can be viewed as attempts to approximate the covariance matrix $\mathbf{\Sigma}$. However, the approximation based on the factor analysis model is more elaborate.
- The primary question in factor analysis is whether the data are consistent with a prescribed structure.

The Orthogonal Factor Model

- The observable random vector $\boldsymbol{X}$, with p components, has mean $\boldsymbol{\mu}$ and covariance matrix Σ.
- The factor model postulates that $\boldsymbol{X}$ is linearly dependent upon a few unobservable random variables $F_1, F_2, \ldots, F_m$, called common factors, and p additional sources of variation $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_p$, called errors, sometimes, specific factors.

In particular, the factor analysis model is

$$
\begin{aligned}
X_1 - \mu_1 &= \ell_{11}F_1 + \ell_{12}F_2 + \cdots + \ell_{1m}F_m + \varepsilon_1 \\
X_2 - \mu_2 &= \ell_{21}F_1 + \ell_{22}F_2 + \cdots + \ell_{2m}F_m + \varepsilon_1 \\
&\vdots \\
X_p - \mu_p &= \ell_{p1}F_1 + \ell_{p2}F_2 + \cdots + \ell_{pm}F_m + \varepsilon_p
\end{aligned}
$$

or in matrix notation

$$\boldsymbol{X} - \boldsymbol{\mu} = \mathbf{LF} + \boldsymbol{\varepsilon}$$

The coefficient ℓ_{ij} is called the loading of the i th variable on the j th factor, so the matrix $\mathbf{L}$ is the matrix of factor loadings.

The unobservable random vectors $\mathbf{F}$ and ε satisfy the following conditions:

- $\mathbf{F}$ and ε are independent
- $\mathrm{E}(\mathbf{F}) = 0, \mathrm{Cov}(\mathbf{F}) = \mathbf{I}$
- $\mathrm{E}(\boldsymbol{\varepsilon}) = 0, \mathrm{Cov}(\boldsymbol{\varepsilon}) = \boldsymbol{\Psi}$, where $\boldsymbol{\Psi}$ is diagonal matrix.

Covariance structure for the Orthogonal Factor Model

1. $\text{Cov}(\boldsymbol{X}) = \mathbf{L}\mathbf{L}' + \boldsymbol{\Psi}$ or

$$\begin{aligned} \text{Var}\left(X_i\right) &= \ell_{i1}^2 + \cdots + \ell_{im}^2 + \psi_i \hat{=} h_i^2 + \psi_i \\ \text{Cov}\left(X_i, X_k\right) &= \ell_{i1}\ell_{k1} + \cdots + \ell_{im}\ell_{km} \end{aligned}$$

2. $\text{Cov}(\boldsymbol{X}, \mathbf{F}) = \mathbf{L}$ or

$$\text{Cov}\left(X_i, F_j\right) = \ell_{ij}$$

Example

Consider the covariance matrix

$$\mathbf{\Sigma} = \begin{bmatrix} 19 & 30 & 2 & 12 \\ 30 & 57 & 5 & 23 \\ 2 & 5 & 38 & 47 \\ 12 & 23 & 47 & 68 \end{bmatrix}$$

Verifying the relation $\mathbf{\Sigma} = \mathbf{L}\mathbf{L}' + \mathbf{\Psi}$ for two factors

Unfortunately, for the factor analyst, most covariance matrices cannot be factored as $\mathbf{L}\mathbf{L}' + \mathbf{\Psi}$, where the number of factors m is much less than p.

- Example: Let $p = 3$ and $m = 1$, and suppose the random variables X_1, X_2 and X_3 have the positive definite covariance matrix

$$\mathbf{\Sigma} = \begin{bmatrix} 1 & .9 & .7 \\ .9 & 1 & .4 \\ .7 & .4 & 1 \end{bmatrix}$$

  Show $\mathbf{\Sigma}$ can not be factored by a factor analysis model with $m = 1$.

Why ?

- Over-determined Systems: For large p and small m, there are far more unique off-diagonal elements ($p(p-1)/2$) than there are factor loadings (pm). It is rare for a real-world matrix to satisfy all these constraints perfectly.
- Constraint Violations: Even when p is small (as in this example where the number of parameters equals the number of equations), the resulting parameters may not fall within valid ranges (e.g., correlations must be between -1 and 1, and variances must be ≥ 0).
- Incompatibility of Correlations: In this example, X_1 is very strongly correlated with X_2 (0.9) and X_3 (0.7). However, X_2 and X_3 are only weakly correlated with each other (0.4). A single common factor cannot simultaneously explain why X_1 is "close" to both variables while those variables remain "far" from each other.

Factor loadings $\mathbf{L}$ are determined only up to an orthogonal matrix $\mathbf{T}$. Thus, the loadings

$$\mathbf{L}^* = \mathbf{L}\mathbf{T} \quad \text{and} \quad \mathbf{L}$$

both give the same representation. The communalities, given by the diagonal elements of $\mathbf{L}'\mathbf{L} = (\mathbf{L}^*)(\mathbf{L}^*)'$ are also unaffected by the choice of $\mathbf{T}$.

$$\boldsymbol{X} - \boldsymbol{\mu} = \mathbf{L}\mathbf{F} + \boldsymbol{\varepsilon} = \mathbf{L}\mathbf{T}\mathbf{T}'\mathbf{F} + \boldsymbol{\varepsilon} = \mathbf{L}^*\mathbf{F}^* + \boldsymbol{\varepsilon}$$

$$\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}' + \boldsymbol{\Psi} = \mathbf{L}\mathbf{T}\mathbf{T}'\mathbf{L}' + \boldsymbol{\Psi} = (\mathbf{L}^*)(\mathbf{L}^*)' + \boldsymbol{\Psi}$$

The principal component analysis of the sample covariance matrix $\mathbf{S}$ is specified in terms of its eigenvalue-eigenvector pairs $\left(\hat{\lambda}_1, \hat{\mathbf{e}}_1\right), \left(\hat{\lambda}_2, \hat{\mathbf{e}}_2\right), \ldots, \left(\hat{\lambda}_p, \hat{\mathbf{e}}_p\right)$, where $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_p$. Let $m < p$ be the number of common factors. Then the matrix of estimate factor loading $\left\{\tilde{\ell}_{ij}\right\}$ is give by

$$\tilde{\mathbf{L}} = \left[\sqrt{\hat{\lambda}_1}\hat{\mathbf{e}}_1 : \sqrt{\hat{\lambda}_2}\hat{\mathbf{e}}_2 : \cdots : \sqrt{\hat{\lambda}_m}\hat{\mathbf{e}}_m\right]$$

The estimate specific variances are provided by the diagonal elements of the matrix $\mathbf{S} - \tilde{\mathbf{L}}\tilde{\mathbf{L}}'$, so

$$\boldsymbol{\Psi} = \begin{bmatrix} \tilde{\psi}_1 & 0 & \cdots & 0 \\ 0 & \tilde{\psi}_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \tilde{\psi}_p \end{bmatrix} \quad \text{with} \quad \tilde{\psi}_i = s_{ii} - \sum_{j=1}^{m} \tilde{\ell}_{ij}^2$$

Communalities are estimated as

$$\tilde{h}_i^2 = \tilde{\ell}_{i1}^2 + \tilde{\ell}_{i2}^2 + \cdots + \tilde{\ell}_{im}^2$$

The principal component factor analysis of the sample correlation matrix is obtained by starting with $\mathbf{R}$ in place of $\mathbf{S}$.

- For the principal component solution, the estimated loading for a given factor do not changes as the number of factors is increased.
- The choice of m can be based on the estimated eigenvalues in much the same manner as with principal components.
- Analytically, we have

$$\text{Sum of squareed entries of } \left(\mathbf{S} - \left(\tilde{\mathbf{L}}\tilde{\mathbf{L}}' + \tilde{\boldsymbol{\Psi}}\right)\right) \leq \hat{\lambda}_{m+1}^2 + \cdots + \hat{\lambda}_p^2$$

- Ideally, the contributions of the first few factors to the sample variance of the variables should be large.

$$\left(\begin{array}{c}\text{Proportion of total}\\ \text{sample variance}\\ \text{due to } j \text{ th factor}\end{array}\right) = \begin{cases} \dfrac{\hat{\lambda}_j}{s_{11}+s_{22}+\cdots+s_{pp}} & \text{for a factor analysis of } \mathbf{S} \\ \dfrac{\lambda_j}{p} & \text{for a factor analysis of } \mathbf{R} \end{cases}$$

Example

In a consumer-preference study, a random sample of customers were asked to rate several attributions of a new product. The response, on a 7 -point semantic differential scale, were tabulated and the attribute correlation matrix constructed. The correlation matrix is presented next:

Attribute (Variable)		1	2	3	4	5
Taste	1	1.00	.02	.96	.42	.01
Good buy for money	2	.02	1.00	.13	.71	.85
Flavor	3	.96	.13	1.00	.50	.11
Suitable for snack	4	.42	.71	.50	1.00	.79
Provides lots of energy	5	.01	.85	.11	.79	1.00

do factor analysis for this consumer-preference data

Variable	Estimated factor loadings $\widetilde{\ell}_{ij} = \sqrt{\hat{\lambda}_i} e_{ij}$		Communalities $\widetilde{h}_i^2$	Specific variances $\widetilde{\psi}_i = 1 - \widetilde{h}_i^2$
	F_1	F_2		
1. Taste	.56	.82	.98	.02
2. Good buy for money	.78	-.53	.88	.12
3. Flavor	.65	.75	.98	.02
4. Suitable for snack	.94	-.10	.89	.11
5. Provides lots of energy	.80	-.54	.93	.07
Eigenvalues	2.85	1.81		
Cumulative proportion of total (standardized) sample variance	.571	.932		

Example

Stock-price data consisting of $n = 103$ weekly rates of return on $p = 5$ stocks. Do factor analysis for this data.

Variable	One-factor solution		Two-factor solution		
	Estimated factor loadings	Specific variances	Estimated factor loadings		Specific variances
	F_1	$\widetilde{\psi}_i = 1 - \widetilde{h}_i^2$	F_1	F_2	$\widetilde{\psi}_i = 1 - \widetilde{h}_i^2$
1. J P Morgan	.732	.46	.732	-.437	.27
2. Citibank	.831	.31	.831	-.280	.23
3. Wells Fargo	.726	.47	.726	-.374	.33
4. Royal Dutch Shell	.605	.63	.605	.694	.15
5. ExxonMobil	.563	.68	.563	.719	.17
Cumulative proportion of total (standardized) sample variance explained	.487		.487	.769	

Results

Let $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_n$ be a random sample from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}' + \boldsymbol{\Psi}$ is the covariance matrix for the m common factor model. The maximum likelihood estimator $\hat{\mathbf{L}}$ and $\hat{\boldsymbol{\Psi}}$ and $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$ maximize the likelihood function of $\boldsymbol{X}_j - \boldsymbol{\mu} = \mathbf{L}\mathbf{F}_j + \varepsilon_j, j = 1, 2, \ldots, n$

$$L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = (2\pi)^{-\frac{np}{2}} |\boldsymbol{\Sigma}|^{-\frac{n}{2}} e^{-\frac{1}{2}\operatorname{tr}\left[\boldsymbol{\Sigma}^{-1}\left(\sum_{j=1}^{n}\left(\mathbf{x}_j - \bar{\mathbf{x}}\right)\left(\mathbf{x}_j - \bar{\mathbf{x}}\right)' + n(\bar{\mathbf{x}} - \boldsymbol{\mu})(\bar{\mathbf{x}} - \boldsymbol{\mu})'\right)\right]}$$

subject to $\hat{\mathbf{L}}\hat{\boldsymbol{\Psi}}^{-1}\hat{\mathbf{L}}$ be diagonal.

The maximum likelihood estimates of the communalities are

$$\hat{h}_i^2 = \hat{\ell}_{i1}^2 + \hat{\ell}_{i2}^2 + \cdots + \hat{\ell}_{im}^2 \text{ for } i = 1, 2, \ldots, p$$

$$\left(\begin{array}{c} \text{Proportion of total sample} \\ \text{variance due to } j \text{ th factor} \end{array} \right) = \frac{\hat{\ell}_{1j}^2 + \hat{\ell}_{2j}^2 + \cdots + \hat{\ell}_{pj}^2}{s_{11} + s_{22} + \cdots + s_{pp}}$$

- Although the likelihood in this Results is appropriate for $\mathbf{S}$, not $\mathbf{R}$, surprisingly, this practice is equivalent to obtaining the maximum likelihood estimate $\hat{\mathbf{L}}$ and $\hat{\boldsymbol{\Psi}}$ based on the sample covariance matrix $\mathbf{S}$, setting $\hat{\mathbf{L}}_z = \hat{\mathbf{V}}^{-1/2}\hat{\boldsymbol{\Psi}}\hat{\mathbf{V}}^{-1/2}$.
- Here $\hat{\mathbf{V}}^{-1/2}$ is the diagonal matrix with reciprocal of the sample standard deviation (computed with the divisor $\sqrt{n}$) on the main diagonal, and $\mathbf{Z}$ is the standardized observation with sample mean 0 and sample standard deviation 1 .

Example

Using the maximum likelihood method do factor analysis for the stock-price data.

Variable	Maximum likelihood			Principal components		
	Estimated factor loadings		Specific variances	Estimated factor loadings		Specific variances
	F_1	F_2	$\hat{\psi}_i = 1 - \hat{h}_i^2$	F_1	F_2	$\tilde{\psi}_i = 1 - \tilde{h}_i^2$
1. J P Morgan	.322	.755	.42	.732	-.437	.27
2. Citibank	.182	.652	.27	.831	-.280	.23
3. Wells Fargo	1.000	-.000	.00	.726	-.374	.33
4. Royal Dutch Shell	.683	-.032	.53	.563	.694	.15
5. Texaco						
Cumulative proportion of total (standardized) sample variance explained	.323	.647		.487	.769	

Example: Factor analysis of Olympic decathlon data)

Linden originally conducted a factor analytic study of Olympic decathlon results for all 160 complete starts from the end of World War II until the mid-seventies . Following his approach we examine the $n = 280$ complete starts from 1960 through 2004. The recorded values for each event were standardized and the signs of the timed events changed so that large scores are good for all events. We, too, analyze the correlation matrix, which based on all 280 cases.

	Principal component					Maximum likelihood				
Variable	Estimated factor loadings				Specific variances	Estimated factor loadings				Specific variances
	F_1	F_2	F_3	F_4		F_1	F_2	F_3	F_4	
1. 100 -m run	.696	.022	-.468	-.416	.12	.993	-.069	-.021	.002	.01
2. Long jump	.793	.075	-.255	-.115	.29	.665	.252	.239	.220	.39
3. Shot put	.771	-.434	.197	-.112	.17	.530	.777	-.141	-.079	.09
4. High jump	.711	.181	.005	.367	.33	.363	.428	.421	.424	.33
5. 400 $-$ m run	.605	.549	-.045	-.397	.17	.571	.019	.620	-.305	.20
6. 100 m hurdles	.513	-.083	-.372	.561	.28	.343	189	.090	.323	.73
7. Discus	.690	-.456	.289	-.078	.23	.402	.718	-.102	-.095	.30
8. Pole vault	.761	.162	.018	.304	.30	.440	.407	.390)	.263	.42
9. Javelin	.518	-.252	.519	-.074	.39	.218	.461	.084	-.085	.73
10. 1500 $-$ m run	.220	.746	.493	.085	.15	-.016	.091	.609	-.145	.60
Cumulative proportion of total variance explained	.42	.56	.67	.76		.27	.45	.57	.62	

$$\mathbf{R} - \widetilde{\mathbf{L}}\widetilde{\mathbf{L}}' - \widetilde{\Psi} =$$

$$\begin{bmatrix}
0 & -.082 & -.006 & -.021 & -.068 & .031 & -.016 & .003 & .039 & .062 \\
-.082 & 0 & -.046 & .033 & -.107 & -.078 & -.048 & -.059 & .042 & .006 \\
-.006 & -.046 & 0 & .006 & -.010 & -.014 & -.003 & -.013 & -.151 & .055 \\
-.021 & .033 & .006 & 0 & -.038 & -.204 & -.015 & -.078 & -.064 & -.086 \\
-.068 & -.107 & -.010 & -.038 & 0 & .096 & .025 & -.006 & .030 & -.074 \\
.031 & -.078 & -.014 & -.204 & .096 & 0 & .015 & -.124 & .119 & .085 \\
-.016 & -.048 & -.003 & -.015 & .025 & .015 & 0 & -.029 & -.210 & .064 \\
.003 & -.059 & -.013 & -.078 & -.006 & -.124 & -.029 & 0 & -.026 & -.084 \\
.039 & .042 & -.151 & -.064 & .030 & .119 & -.210 & -.026 & 0 & -.078 \\
.062 & .006 & .055 & -.086 & -.074 & .085 & .064 & -.084 & -.078 & 0
\end{bmatrix}$$

Factor Rotation

If $\hat{\mathbf{L}}$ is the $p \times m$ matrix of estimated factor loadings obtained by any method (principal component,maximum likelihood, and so forth) then

$$\hat{\mathbf{L}}^* = \mathbf{L}\mathbf{T}, \text{ where } \mathbf{T}\mathbf{T}' = \mathbf{T}'\mathbf{T} = \mathbf{I}$$

is a $p \times m$ matrix of "rotated" loadings. Moreover, the estimated covariance (or correlation) matrix remains unchanged, since

$$\hat{\mathbf{L}}\hat{\mathbf{L}}' + \hat{\mathbf{\Psi}} = \hat{\mathbf{L}}\mathbf{T}\mathbf{T}'\hat{\mathbf{L}}' + \hat{\mathbf{\Psi}} = \hat{\mathbf{L}}^*\hat{\mathbf{L}}^{*\prime} + \hat{\mathbf{\Psi}}$$

Example (A first look at factor rotation)

Lawley and Maxwell present the sample correlation matrix of examination scores in $p = 6$ subject areas for $n = 220$ male students. The correlation matrix is

$$\mathbf{R} = \begin{array}{c} \begin{array}{cccccc} \text{Gaelic} & \text{English} & \text{History} & \text{Arithmetic} & \text{Algebra} & \text{Geometry} \end{array} \\ \begin{bmatrix} 1.0 & .439 & .410 & .288 & .329 & .248 \\ & 1.0 & .351 & .354 & .320 & .329 \\ & & 1.0 & .164 & .190 & .181 \\ & & & 1.0 & .595 & .470 \\ & & & & 1.0 & .464 \\ & & & & & 1.0 \end{bmatrix} \end{array}$$

and a maximum likelihood solution for $m = 2$ common factors yields the estimates in Table 9.5

Variable	Estimated factor loadings		Communalities $\hat{h}_i^2$
	F_1	F_2	
1. Gaelic	.553	.429	.490
2. English	.568	.288	.406
3. History	.392	.450	.356
4. Arithmetic	.740	-.273	.623
5. Algebra	.724	-.211	.569
6. Geometry	.595	-.132	.372

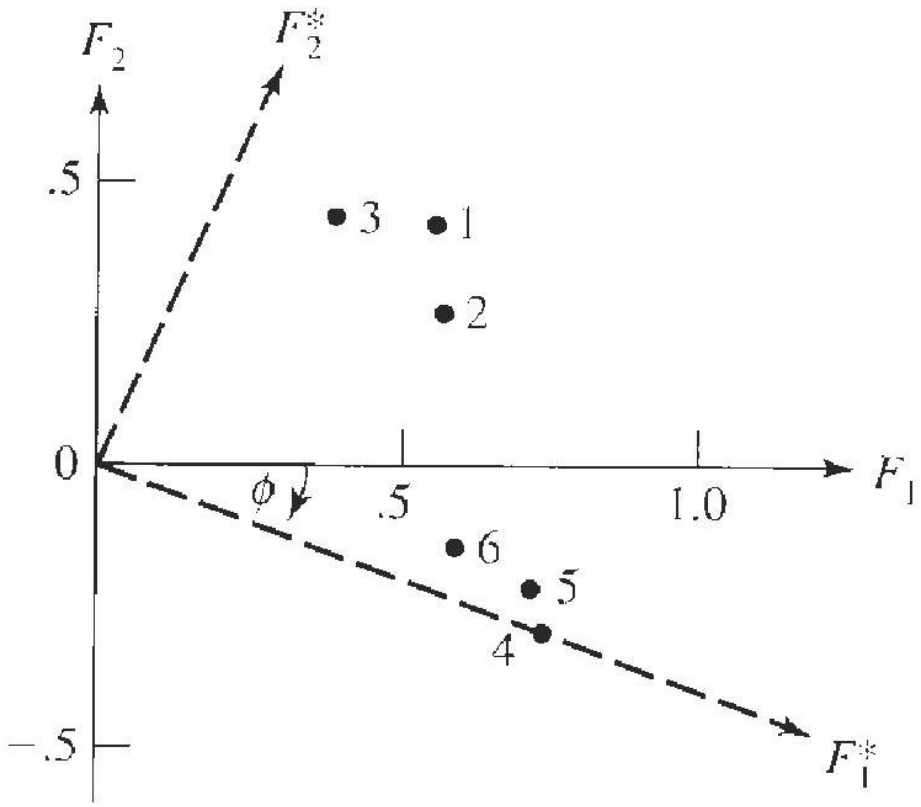


Figure 9.1 Factor rotation for test scores.

- Define $\tilde{\ell}^*_{ij} = \hat{\ell}^*_{ij}/\hat{h}_i$ to be the rotated coefficients scaled by the square root of the communalities. Then the (normal) varimax procedure selects the orthogonal transformation $\mathbf{T}$ that makes

$$V = \frac{1}{p}\sum_{j=1}^{m}\left[\sum_{i=1}^{p}\tilde{\ell}^{*4}_{ij} - \left(\sum_{i=1}^{p}\tilde{\ell}^{*2}_{ij}\right)^2 / p\right]$$

  as large as possible.
- Scaling the rotated coefficient $\hat{\ell}^*_{ij}$ has the effect of giving variables with small communalities relatively more weight in the determination of simple structure. After the transformation $\mathbf{T}$ is determined, the loadings $\tilde{\ell}^*_{ij}$ are multiplied by $\hat{h}_i$ so that the original communalities are preserved.

Equalizing the Influence of Variables

In factor analysis, some variables may have very high communalities ($\hat{h}_i^2$), while others have very low ones.

- If we used the raw loadings $\hat{\ell}_{ij}$ to maximize V, the variables with the largest communalities would dominate the optimization process.
- By dividing by $\hat{h}_i$ (the square root of the communality), we project the loadings onto a unit sphere. This gives every variable "equal weight" during the rotation, ensuring the final factor structure is representative of all variables, not just the ones with high variance.

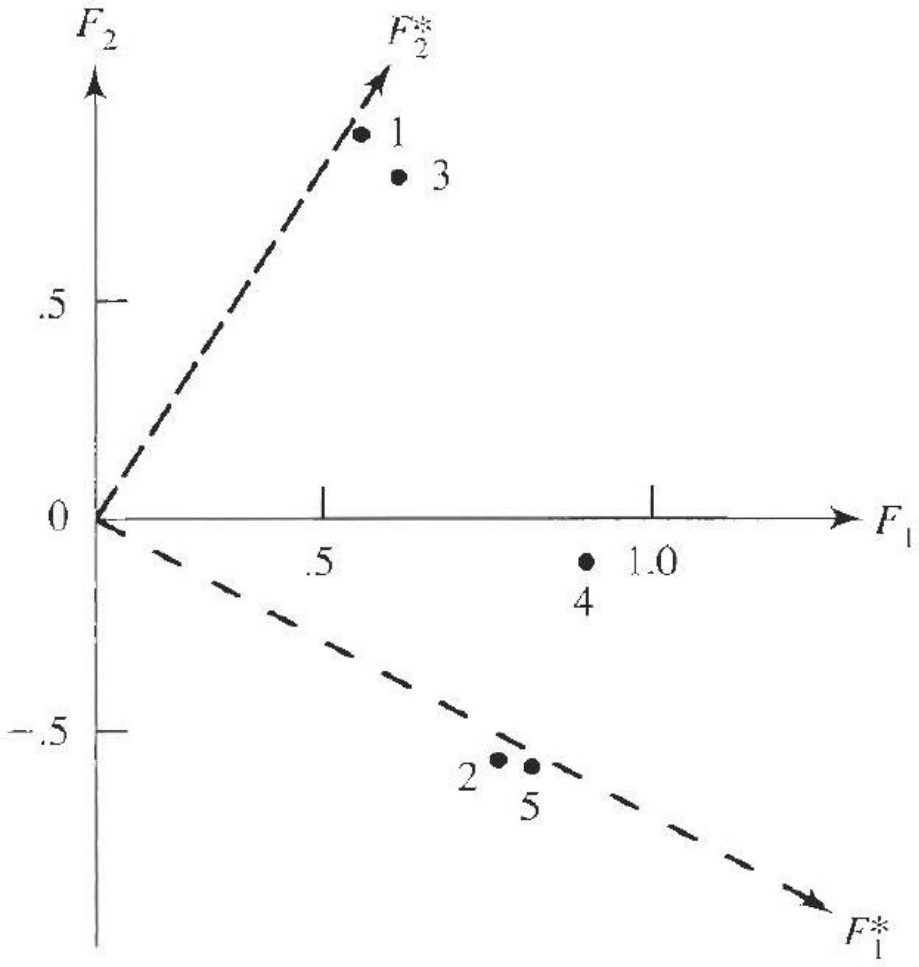
The goal of Varimax is to achieve simple structure. A simple structure means each variable loads highly on only one factor and near zero on others.

- The formula for V is essentially the sum of the variances of the squared loadings within each factor:

$$V = \frac{1}{p} \sum_{j=1}^{m} \mathrm{Var}(\tilde{\ell}_{1j}^{*2}, \ldots, \tilde{\ell}_{pj}^{*2})$$

  Maximizing Variance: To make the variance of the squared loadings as large as possible, you need the values to be at the extremes—either very large (close to 1) or very small (close to 0).
- The Result: This "pushes" the loadings away from the middle ground, making the factors much easier to interpret because the associations between variables and factors become binary (yes/no) rather than ambiguous.

Example (Rotated Loading for the consumer-preference data)



Rotation Method Varimax		
	Rotated Factor Pattern	
	FACTOR 1	FACTOR2
TASTE	0.01970	0.98948
MONEY	0.93744	-0.01123
FLAVOR	0.12856	0.97947
SNACK	0.84244	0.42805
ENERGY	0.96539	-0.01563
Variance explained by each factor		
	FACTOR1	FACTOR2
	2.537396	2.122027

	Maximum likelihood estimates of factor loadings		Rotated estimated factor loadings		Specific variances $\hat{\psi}_i^2 = 1 - \hat{h}_i^2$
	F_1	F_2	F_1^*	F_2^*	
JP Morgan	.115	.755	.763	.024	.42
Citibank	.322	.788	.821	.227	.27
Wells Fargo	.182	.652	.669	.104	.54
Royal Dutch Shell	1.000	-.000	.118	.993	.00
ExxonMobil	.683	.032	.113	.675	.53
Cumulative proportion of total sample variance explained	.323	.647	.346	.647	

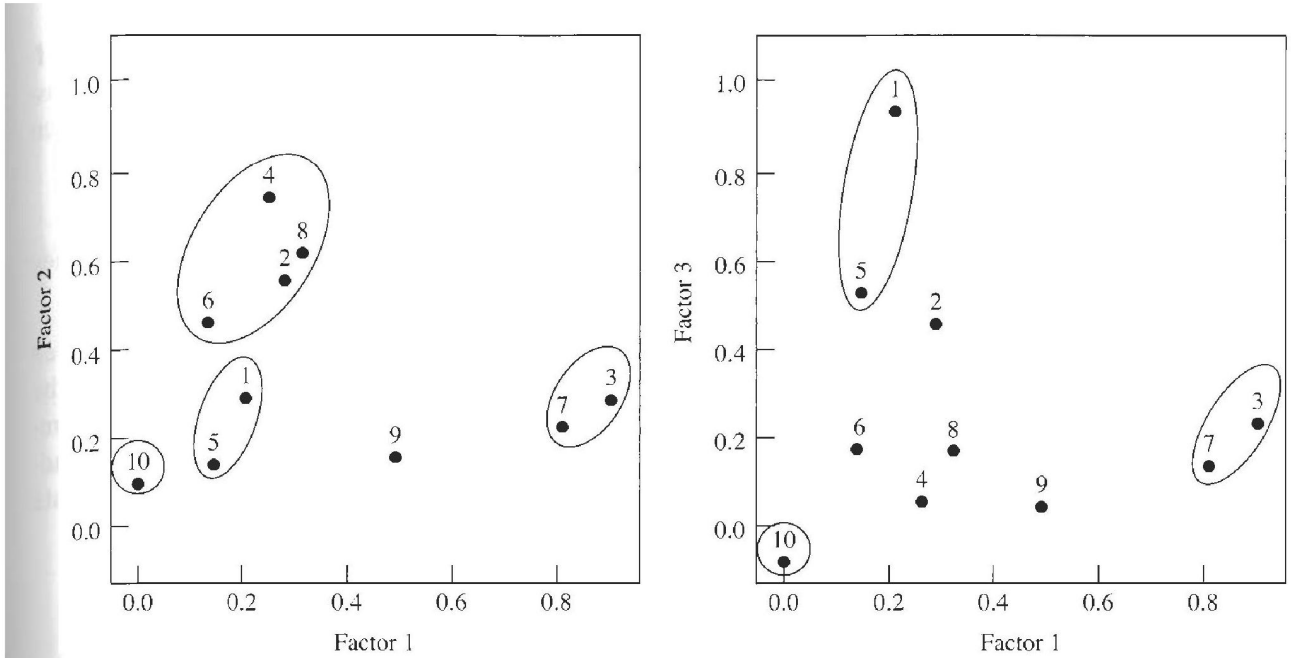


Figure 9.3 Rotated maximum likelihood loadings for factor pairs $(1, 2)$ and $(1, 3)$ decathlon data. (The numbers in the figures correspond to variables.)

Factor Scores

- The estimate values of the common factors, called factor scores may also required. These quantities are often used for diagnostic purposes, as well as inputs to a subsequent analysis.
- Factor scores are not estimates of unknown parameters in the usual sense. Rather, they are estimates of values for the unobserved random factor vectors $\mathbf{F}_j, j = 1, 2, \ldots, n$. That is, factor scores

$$\hat{\mathbf{f}}_j = \text{ estimate of the values } \mathbf{f}_j \text{ attained by } \mathbf{F}_j \text{ (} j \text{ th case)}$$

- Normally the factor score approaches have two elements in common:

1. They treat the estimate factor loadings $\hat{\ell}_{ij}$ and specific variance $\hat{\psi}_i$ as if they were true values.
2. They involve linear transformations of the original data, perhaps centered or standardized. Typically, the estimated rotated loadings, rather than the original estimated loadings, are used to compute factor scores.

Factor Scores Obtained by Weighted Least Squares from the Maximum Likelihood Estimates

$$\begin{aligned}\hat{\mathbf{f}}_j &= \left(\hat{\mathbf{L}}'\hat{\boldsymbol{\Psi}}^{-1}\hat{\mathbf{L}}\right)^{-1}\hat{\mathbf{L}}'\hat{\boldsymbol{\Psi}}^{-1}\left(\mathbf{x}_j - \hat{\boldsymbol{\mu}}\right)\\ &= \hat{\boldsymbol{\Delta}}^{-1}\hat{\mathbf{L}}'\hat{\boldsymbol{\Psi}}^{-1}\left(\mathbf{x}_j - \hat{\mathbf{x}}\right), j = 1, 2, \ldots, n\end{aligned}$$

or, if the correlation matrix is factored

$$\begin{aligned}\hat{\mathbf{f}}_j &= \left(\hat{\mathbf{L}}_z'\hat{\boldsymbol{\Psi}}_z^{-1}\hat{\mathbf{L}}_z\right)^{-1}\hat{\mathbf{L}}_z'\hat{\boldsymbol{\Psi}}_z^{-1}\mathbf{z}_j\\ &= \hat{\boldsymbol{\Delta}}_z^{-1}\hat{\mathbf{L}}_z'\hat{\boldsymbol{\Psi}}_z^{-1}\mathbf{z}_j, j = 1, 2, \ldots, n\end{aligned}$$

where $\mathbf{z}_j = \mathbf{D}^{-1/2}\left(\mathbf{x}_j - \overline{\mathbf{x}}\right)$ and $\hat{\boldsymbol{\rho}} = \hat{\mathbf{L}}_z\hat{\mathbf{L}}_z' + \hat{\boldsymbol{\Psi}}_z$.

- If rotated loadings $\hat{\mathbf{L}}^* = \hat{\mathbf{L}}\mathbf{T}$ are used in place of the original loadings, the subsequent factor scores, $\hat{\mathbf{f}}_j^*$, are related to $\hat{\mathbf{f}}_j$ by $\hat{\mathbf{f}}_j^* = \mathbf{T}'\hat{\mathbf{f}}_j, j = 1, 2, \ldots, n$.
- If the factor loadings are estimated by the principal component method, it is customary to generate factor scores using an unweighted (ordinary) least squares procedure. Implicitly, this amount to assuming that the ψ_i are equal or nearly equal. The factor scores are then

$$\hat{\mathbf{f}}_j = \left(\hat{\mathbf{L}}'\hat{\mathbf{L}}\right)^{-1}\hat{\mathbf{L}}'\left(\mathbf{x}_j - \hat{\boldsymbol{\mu}}\right) \quad \text{or} \quad \hat{\mathbf{f}}_j = \left(\hat{\mathbf{L}}_z'\hat{\mathbf{L}}_z\right)^{-1}\hat{\mathbf{L}}_z'\mathbf{z}_j$$

for standardized data.

$$\hat{\mathbf{f}}_j = \hat{\mathbf{L}}'\mathbf{S}^{-1}\left(\mathbf{x}_j - \overline{\mathbf{x}}\right), j = 1, 2, \ldots, n$$

or, if a correlation matrix is factored

$$\hat{\mathbf{f}}_j = \hat{\mathbf{L}}_z'\mathbf{R}^{-1}\mathbf{z}_j, j = 1, 2, \ldots, n$$

where $\mathbf{z}_j = \mathbf{D}^{-1/2}\left(\mathbf{x}_j - \overline{\mathbf{x}}\right)$ and $\hat{\boldsymbol{\rho}} = \hat{\mathbf{L}}_z\hat{\mathbf{L}}_z' + \hat{\boldsymbol{\Psi}}_z$.

Example (Computing factor scores)

Compute factor scores by the least squares and regression methods using the stock-price data.

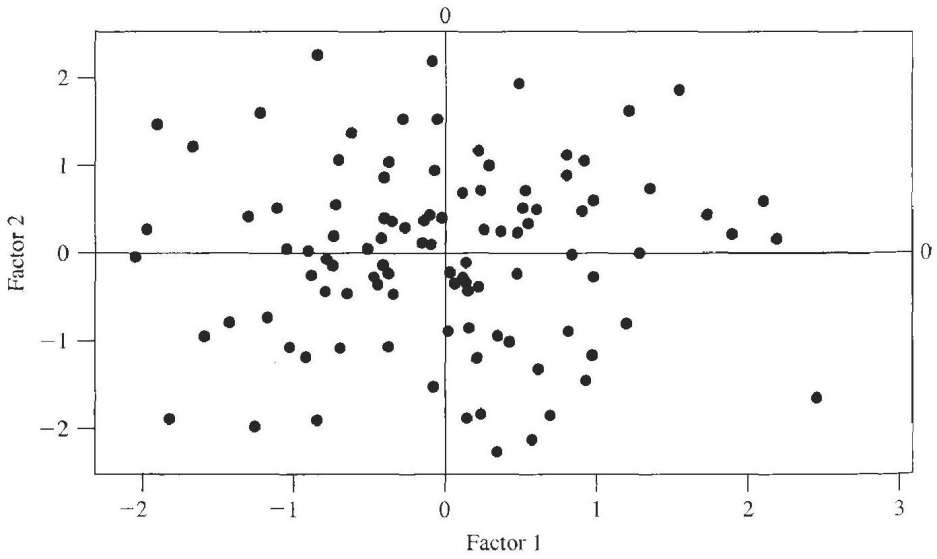


Figure 9.4 Factor scores for factors 1 and 2 of the stock-price data (maximum likelihood estimates of the factor loadings).

Perspectives and a Strategy for Factor Analysis

At the present time, factor analysis still maintains the flavor of an art, and no single strategy should yet be "chiseled into stone". We suggest and illustrate one reasonable option:

1. Perform a principal component factor analysis. This method is particularly appropriate for a first pass through the data. (It is not required that **R** or **S** be nonsingular)
    1. Look for suspicious observations by plotting the factor scores. Also, calculate standardized scores for each observation and squared distances.
    2. Try a varimax rotation.

2. Perform a maximum likelihood factor analysis, including a varimax rotation.
3. Compare the solution obtained from the two factor analysis.
    1. Do the loadings group in the same manner?
    2. Plot factor scores obtained for principal components against scores from the maximum likelihood analysis.

4. Repeat the first three steps for other numbers of common factors m. Do extra factors necessarily contribute to the understanding and interpretation of the data?
5. For large data sets, split them in half and perform a factor analysis on each part. Compare the two results with each other and with that obtained from the complete data set to check the stability of the solution.

Summary: Core Concepts

While both factor analysis and PCA techniques reduce the number of variables, their **objectives** are fundamentally different:

- **PCA**: Focuses on explaining the **Total Variance**. It is a data reduction tool used to summarize information into fewer variables.
- **Factor Analysis**: Focuses on explaining the **Covariance (Correlations)** between variables. It seeks to identify underlying "hidden" constructs.

Direction of Causality

The primary difference lies in how the components/factors are related to the observed variables.

PCA: Bottom-Up

Components are **linear combinations** of the observed variables.

$$PC_i = a_{i1}X_1 + a_{i2}X_2 + \cdots + a_{ip}X_p$$

Factor Analysis: Top-Down

Observed variables are **caused by** the underlying factors plus error.

$$X_i = l_{i1}F_1 + l_{i2}F_2 + \cdots + \epsilon_i$$

Variance Partitioning

Factor Analysis divides variance into two parts, whereas PCA treats it as a single block:

1. **Common Variance (Communality)**: The portion of variance shared with other variables.
2. **Unique Variance (Specificity + Error)**: Variance specific to that variable and measurement error.

- **PCA**: Uses 100% of the variance (Common + Unique).
- **FA**: Models *only* the Common Variance.

Comparison Summary

Feature	PCA	Factor Analysis
Goal	Data reduction	Structure discovery
Variance analyzed	Total Variance	Common Variance
Logic	$X \to$ Component	Factor $\to X$
Error Term	Not modeled	Explicitly modeled (ϵ)
Rotation	Less common	Standard (Varimax, etc.)

When to Use Which?

- **Use PCA if...**
  - You want to reduce 50 variables to 5 for a regression model.
  - You don't care about "naming" the components.
  - You assume your variables are measured without error.

- **Use Factor Analysis if...**
  - You want to understand why variables correlate (e.g., "Intelligence", "Anxiety").
  - You want to identify latent constructs for theory building.
  - You want to account for measurement error.

A Note on Rotation

- In **Factor Analysis**, we often "rotate" factors to achieve **Simple Structure**.
- **Varimax (Orthogonal)**: Keeps factors uncorrelated; easier to interpret.
- **Promax (Oblique)**: Allows factors to correlate; often more realistic for human behaviors/traits.